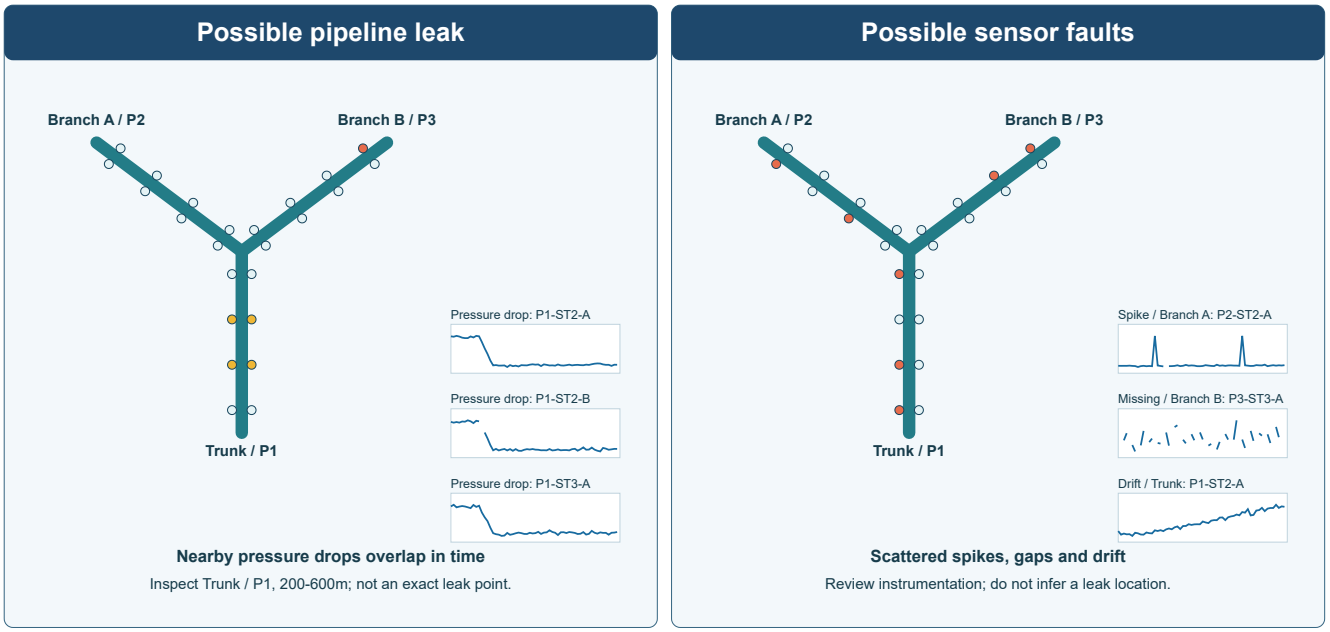
Subsea Integrity Copilot - Customer Brief

Prospect and problem. NorthSea Integrity Operations is a fictional offshore gas operator. Engineers must distinguish pipeline leaks from sensor spikes, gaps and drift. Missed leaks threaten the environment and operations; false alarms trigger costly inspections. Scarce expertise is difficult to encode and transfer to newcomers.



Existing detector alerts → Agent-assisted investigation → Human review

Patterns are evidence, not proof: check operating changes and shared failures.

Demo assumptions: 24 pipe-mounted pressure instruments, 12 paired stations, three connected 800m segments.

Same pressure-pack-v3 geometry as console; seed 2026, [80,140)s, pressure in bar; independent trace y-scales; synthetic.

Why this problem. My PhD studied marine data quality, anomaly detection and translating expertise into reviewable smart systems.

The two customer roles. The **Head of Subsea Operations** owns the budget: cost, workload, timeline and risk. The **Lead Subsea Integrity & Instrumentation Engineer** challenges calibration, shared failures, missing data and unsupported conclusions. Engineers and newcomers use the copilot.

Why unsigned. Doubts about effort savings, hallucinations, missed leaks, integration cost, privacy and accountability. Cheaper fixed workflows are credible alternatives.

Monitoring workloads. Original Y: three 800m pipes, 24 instruments. Extended: seven branches, 11.2km, 112 instruments; overlapping near/far responses and faults. Median/MAD feeds sliding-window screening; agents compare joint patterns, records and failure dependencies. Ambiguous locations are withheld. Questions: "Leak or faulty sensors?", "Where should we inspect, and why?" and "What evidence supports this, and what remains unknown?"

Agent value. Three scoped roles investigate quality, choose incident-time history searches/comparisons and explain competing causes. A separate same-model critic reviews original sources; code checks quotes, coverage and identity. Revise once, then accept or withhold. Engineers inspect sources and download evidence. Versioned research-curated guidance is not SME-approved or automatically acquired expertise.

Limits and buying gate. Fictional dirty/conflicting data; unvalidated hydraulics; intervals, not exact positions. Original cohort: hosted 15/18 first attempts, then 3/3 explicit retries; fixed 18/18 and faster. No field-accuracy guarantee. Warnings never wait for inference. Manual time is unmeasured; reports target five minutes, counting failures and human review separately. Propose a four-week read-only shadow pilot with agreed quality, abstention, latency and cost gates. Pricing requires discovery; savings are unproven, connectors unbuilt and hosted data requires approval. Humans retain safety authority. Buy only if quality, effort and total cost beat manual/fixed baselines.

Private GitHub and run instructions: github.com/ntnguyen-so/subsea-integrity-copilot. Repository access must be granted before review.